MAT2002 HW5 文杰 123090612

1

First find the eigenvalues by solving the characteristic equation :

$$\det\begin{pmatrix}1-\lambda & 1 & 1\\ 2 & 1-\lambda & -1\\ -8 & -5 & -3-\lambda\end{pmatrix}=0$$

Expanding the determinant:

$$(1-\lambda)[(1-\lambda)(-3-\lambda)-5]-1[2(-3-\lambda)-8]+1[-10-(-8)(1-\lambda)]=0$$
$$-(\lambda+1)(\lambda^2-4)=0$$

The eigenvalues are $\lambda_1=-1$, $\lambda_2=2$, and $\lambda_3=-2$.

Since the eigenvalues are real and distinct: it's **Case 1** now.

For $\lambda_1=-1$:

$$\begin{pmatrix}2 & 1 & 1\\ 2 & 2 & -1\\ -8 & -5 & -2\end{pmatrix}\begin{pmatrix}\xi_1\\ \xi_2\\ \xi_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}\implies \boldsymbol{\xi}_1=\begin{pmatrix}-3\\ 4\\ 2\end{pmatrix}$$

For $\lambda_2=2$:

$$\begin{pmatrix}-1 & 1 & 1\\ 2 & -1 & -1\\ -8 & -5 & -5\end{pmatrix}\begin{pmatrix}\xi_1\\ \xi_2\\ \xi_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}\implies \boldsymbol{\xi}_2=\begin{pmatrix}0\\ -1\\ 1\end{pmatrix}$$

For $\lambda_3=-2$:

$$\begin{pmatrix}3 & 1 & 1\\ 2 & 3 & -1\\ -8 & -5 & -1\end{pmatrix}\begin{pmatrix}\xi_1\\ \xi_2\\ \xi_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}\implies \boldsymbol{\xi}_3=\begin{pmatrix}-4\\ 5\\ 7\end{pmatrix}$$

$$\mathbf{x}(t)=c_1\boldsymbol{\xi}_1e^{\lambda_1t}+c_2\boldsymbol{\xi}_2e^{\lambda_2t}+c_3\boldsymbol{\xi}_3e^{\lambda_3t}=c_1\begin{pmatrix}-3\\ 4\\ 2\end{pmatrix}e^{-t}+c_2\begin{pmatrix}0\\ -1\\ 1\end{pmatrix}e^{2t}+c_3\begin{pmatrix}-4\\ 5\\ 7\end{pmatrix}e^{-2t}$$

2

(i)

The characteristic equation is $\det(\mathbf{A}-\lambda\mathbf{I})=0$:

$$\det\begin{pmatrix}1-\lambda & 1 & 2\\ 0 & 2-\lambda & 2\\ -1 & 1 & 3-\lambda\end{pmatrix}=0$$

Expanding along the first column:

$$(1-\lambda)[(2-\lambda)(3-\lambda)-2]-1[2-2(2-\lambda)]=0$$
$$-(\lambda-1)(\lambda-2)(\lambda-3)=0$$

The eigenvalues are $\lambda_1=1,\lambda_2=2,\lambda_3=3$.

- **For $\lambda_1=1$:** $(\mathbf{A}-\mathbf{I})\boldsymbol{\xi}=\mathbf{0}\implies\begin{pmatrix}0 & 1 & 2\\ 0 & 1 & 2\\ -1 & 1 & 2\end{pmatrix}\begin{pmatrix}\xi_1\\ \xi_2\\ \xi_3\end{pmatrix}=\mathbf{0}\implies\xi_1=0,\xi_2=-2\xi_3$. Let $\boldsymbol{\xi}_1=\begin{pmatrix}0\\ -2\\ 1\end{pmatrix}$.
- **For $\lambda_2=2$:** $(\mathbf{A}-2\mathbf{I})\boldsymbol{\xi}=\mathbf{0}\implies\begin{pmatrix}-1 & 1 & 2\\ 0 & 0 & 2\\ -1 & 1 & 1\end{pmatrix}\begin{pmatrix}\xi_1\\ \xi_2\\ \xi_3\end{pmatrix}=\mathbf{0}\implies\xi_3=0,\xi_1=\xi_2$. Let $\boldsymbol{\xi}_2=\begin{pmatrix}1\\ 1\\ 0\end{pmatrix}$.
- **For $\lambda_3=3$:** $(\mathbf{A}-3\mathbf{I})\boldsymbol{\xi}=\mathbf{0}\implies\begin{pmatrix}-2 & 1 & 2\\ 0 & -1 & 2\\ -1 & 1 & 0\end{pmatrix}\begin{pmatrix}\xi_1\\ \xi_2\\ \xi_3\end{pmatrix}=\mathbf{0}\implies\xi_1=\xi_2=2\xi_3$. Let $\boldsymbol{\xi}_3=\begin{pmatrix}2\\ 2\\ 1\end{pmatrix}$.

The general solution is $\mathbf{x}(t)=c_1e^t\boldsymbol{\xi}_1+c_2e^{2t}\boldsymbol{\xi}_2+c_3e^{3t}\boldsymbol{\xi}_3$.

At $t=0$: $c_1\begin{pmatrix}0\\ -2\\ 1\end{pmatrix}+c_2\begin{pmatrix}1\\ 1\\ 0\end{pmatrix}+c_3\begin{pmatrix}2\\ 2\\ 1\end{pmatrix}=\begin{pmatrix}2\\ 0\\ 1\end{pmatrix}$

Solving this system: $c_1=1,c_2=2,c_3=0$.

$$\mathbf{x}(t)=e^t\begin{pmatrix}0\\ -2\\ 1\end{pmatrix}+2e^{2t}\begin{pmatrix}1\\ 1\\ 0\end{pmatrix}=\begin{pmatrix}2e^{2t}\\ -2e^t+2e^{2t}\\ e^t\end{pmatrix}$$

Behavior as $t \to \infty$: Since the exponents are positive, the components grow exponentially. The term e^{2t} dominates, and the solution **diverges (blows up)** as $t \to \infty$.

(ii)

$$\det \begin{pmatrix} 1-\lambda & -5 \\ 1 & -3-\lambda \end{pmatrix} = (1-\lambda)(-3-\lambda) + 5 = \lambda^2 + 2\lambda + 2 = 0$$

Using the quadratic formula: $\lambda = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$.

It's a complex conjugate pair with $\delta = -1, \mu = 1$.

For $\lambda = -1 + i$:

$$\begin{pmatrix} 2-i & -5 \\ 1 & -2-i \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \mathbf{0} \implies \xi_1 = (2+i)\xi_2$$

Choosing $\xi_2 = 1$, the complex eigenvector is $\boldsymbol{\xi} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Thus, $\mathbf{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

The real-valued general solution is $\mathbf{x}(t) = e^{\delta t}[c_1(\mathbf{u}\cos \mu t - \mathbf{v}\sin \mu t) + c_2(\mathbf{u}\sin \mu t + \mathbf{v}\cos \mu t)]$.

Applying $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$: $c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \implies c_1 = 1, c_2 = -1$.

Substituting constants back:

$$\mathbf{x}(t) = e^{-t}\left[\begin{pmatrix} 2\cos t - \sin t \\ \cos t \end{pmatrix} - \begin{pmatrix} 2\sin t + \cos t \\ \sin t \end{pmatrix}\right] = e^{-t}\begin{pmatrix} \cos t - 3\sin t \\ \cos t - \sin t \end{pmatrix}$$

Because the real part of the eigenvalues ($\delta = -1$) is negative, the solution involves decaying oscillations. As $t \to \infty$, the solution **spirals inward and decays to the origin** $\mathbf{0}$.

3

For a constant-coefficient homogeneous system, the equation

$$\mathbf{x}'(t) = A\mathbf{x}(t)$$

is solved by finding eigenvalues and eigenvectors; when a complex-conjugate pair $\lambda = \alpha \pm i\beta$ appears, the real-valued general solution is obtained from the real and imaginary parts, giving terms of the form

$$e^{\alpha t}\big(p\cos(\beta t) - q\sin(\beta t)\big), \qquad e^{\alpha t}\big(p\sin(\beta t) + q\cos(\beta t)\big).$$

Let

$$A = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix}.$$

First compute the characteristic polynomial:

$$\det(\lambda I - A) = \begin{vmatrix} \lambda+3 & 0 & -2 \\ -1 & \lambda+1 & 0 \\ 2 & 1 & \lambda \end{vmatrix} = (\lambda+2)(\lambda^2+2\lambda+3).$$

Hence the eigenvalues are

$$\lambda_1 = -2, \qquad \lambda_{2,3} = -1 \pm i\sqrt{2}.$$

For $\lambda_1 = -2$, solving $(A+2I)\mathbf{v} = 0$ gives an eigenvector

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}.$$

So one real solution is

$$\mathbf{x}^{(1)}(t) = e^{-2t}\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}.$$

For $\lambda = -1 + i\sqrt{2}$, one eigenvector is

$$\mathbf{v} = \begin{pmatrix} 2 - i\sqrt{2} \\ -1 - i\sqrt{2} \\ 3 \end{pmatrix} = p + iq,$$

where

$$p = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \qquad q = \begin{pmatrix} -\sqrt{2} \\ -\sqrt{2} \\ 0 \end{pmatrix}.$$

Therefore, from the real-valued form, two linearly independent real solutions are

$$\mathbf{x}^{(2)}(t) = e^{-t}\big(p\cos(\sqrt{2}t) - q\sin(\sqrt{2}t)\big),$$

$$\mathbf{x}^{(3)}(t) = e^{-t}\big(p\sin(\sqrt{2}t) + q\cos(\sqrt{2}t)\big).$$

Writing them out explicitly,

$$\mathbf{x}^{(2)}(t) = e^{-t}\begin{pmatrix} 2\cos(\sqrt{2}t) + \sqrt{2}\sin(\sqrt{2}t) \\ -\cos(\sqrt{2}t) + \sqrt{2}\sin(\sqrt{2}t) \\ 3\cos(\sqrt{2}t) \end{pmatrix},$$

$$\mathbf{x}^{(3)}(t) = e^{-t}\begin{pmatrix} 2\sin(\sqrt{2}t) - \sqrt{2}\cos(\sqrt{2}t) \\ -\sin(\sqrt{2}t) - \sqrt{2}\cos(\sqrt{2}t) \\ 3\sin(\sqrt{2}t) \end{pmatrix}.$$

Hence the general solution in terms of real-valued functions is

$$\boxed{\mathbf{x}(t) = c_1 e^{-2t}\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} 2\cos(\sqrt{2}t) + \sqrt{2}\sin(\sqrt{2}t) \\ -\cos(\sqrt{2}t) + \sqrt{2}\sin(\sqrt{2}t) \\ 3\cos(\sqrt{2}t) \end{pmatrix} + c_3 e^{-t}\begin{pmatrix} 2\sin(\sqrt{2}t) - \sqrt{2}\cos(\sqrt{2}t) \\ -\sin(\sqrt{2}t) - \sqrt{2}\cos(\sqrt{2}t) \\ 3\sin(\sqrt{2}t) \end{pmatrix}}$$

for arbitrary constants $c_1, c_2, c_3 \in \mathbb{R}$.

4

Let

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -8 & -5 & -3 \end{pmatrix}.$$

The eigenvalues and corresponding eigenvectors are

$$\lambda_1 = -2, \quad \xi_1 = \begin{pmatrix} -4 \\ 5 \\ 7 \end{pmatrix}, \qquad \lambda_2 = -1, \quad \xi_2 = \begin{pmatrix} -3 \\ 4 \\ 2 \end{pmatrix}, \qquad \lambda_3 = 2, \quad \xi_3 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Hence one fundamental matrix is

$$\boxed{\Psi(t) = \begin{pmatrix} -4e^{-2t} & -3e^{-t} & 0 \\ 5e^{-2t} & 4e^{-t} & -e^{2t} \\ 7e^{-2t} & 2e^{-t} & e^{2t} \end{pmatrix}}$$

Equivalently,

$$\Psi(t) = Pe^{\Lambda t},$$

where

$$P = \begin{pmatrix} -4 & -3 & 0 \\ 5 & 4 & -1 \\ 7 & 2 & 1 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

The fundamental matrix satisfying $\Phi(0) = I$ is

$$\Phi(t) = \Psi(t)\Psi(0)^{-1} = Pe^{\Lambda t}P^{-1}.$$

Thus

$$\boxed{\Phi(t) = \begin{pmatrix} 3e^{-t} - 2e^{-2t} & e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ \dfrac{3e^{2t} - 8e^{-t} + 5e^{-2t}}{2} & \dfrac{13e^{2t} - 16e^{-t} + 15e^{-2t}}{12} & \dfrac{e^{2t} - 16e^{-t} + 15e^{-2t}}{12} \\ \dfrac{-3e^{2t} - 4e^{-t} + 7e^{-2t}}{2} & \dfrac{-13e^{2t} - 8e^{-t} + 21e^{-2t}}{12} & \dfrac{-e^{2t} - 8e^{-t} + 21e^{-2t}}{12} \end{pmatrix}}$$

Therefore, the general solution can also be written as

$$\mathbf{x}(t) = \Psi(t)\mathbf{c}, \qquad \mathbf{c} \in \mathbb{R}^3.$$

5

(i)

Let $x_1 = u$, let $x_2 = u'$.

Taking the derivative of x_1 & x_2: $x_1' = u' = x_2$, $x_2' = u''$.

Using the original differential equation $u'' + \omega^2 u = 0$:

$$x_2' = -\omega^2 u = -\omega^2 x_1$$

Express this system of two first-order equations in matrix form $\mathbf{x}' = \mathbf{A}\mathbf{x}$:

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

This gives us $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}$$

The initial conditions $u(0) = u_0$ and $u'(0) = v_0$ translate into the initial vector $\mathbf{x}^0$:

$$\mathbf{x}(0) = \begin{pmatrix} u(0) \\ u'(0) \end{pmatrix} = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix} = \mathbf{x}^0$$

(ii)

For any constant square matrix $\mathbf{A}$ and scalar t, the matrix exponential is defined by the convergent power series $\exp(\mathbf{A}t) = \mathbf{I} + \sum_{n=1}^{\infty} \frac{\mathbf{A}^n t^n}{n!}$.

$$\mathbf{A}^2 = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix} = \begin{pmatrix} -\omega^2 & 0 \\ 0 & -\omega^2 \end{pmatrix} = -\omega^2 \mathbf{I}$$

Since $\mathbf{A}^2 = -\omega^2 \mathbf{I}$, establish a pattern for all even and odd powers of the matrix:

- **Even powers:** $\mathbf{A}^{2k} = (-\omega^2)^k \mathbf{I} = (-1)^k \omega^{2k} \mathbf{I}$
- **Odd powers:** $\mathbf{A}^{2k+1} = \mathbf{A}^{2k}\mathbf{A} = (-1)^k \omega^{2k} \mathbf{A}$

Split the Taylor series for $\exp(\mathbf{A}t)$ into its even and odd terms:

$$\exp(\mathbf{A}t) = \sum_{k=0}^{\infty} \frac{\mathbf{A}^{2k} t^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{\mathbf{A}^{2k+1} t^{2k+1}}{(2k+1)!}$$

Substituting our power patterns into the series yields:

$$\exp(\mathbf{A}t) = \sum_{k=0}^{\infty} \frac{(-1)^k \omega^{2k} t^{2k}}{(2k)!} \mathbf{I} + \sum_{k=0}^{\infty} \frac{(-1)^k \omega^{2k} t^{2k+1}}{(2k+1)!} \mathbf{A}$$

By adjusting the terms slightly, match the standard Maclaurin series expansions for cosine and sine:

$$\exp(\mathbf{A}t) = \left(\sum_{k=0}^{\infty} \frac{(-1)^k (\omega t)^{2k}}{(2k)!} \right) \mathbf{I} + \frac{1}{\omega} \left(\sum_{k=0}^{\infty} \frac{(-1)^k (\omega t)^{2k+1}}{(2k+1)!} \right) \mathbf{A}$$

Because the series on the left is the expansion for $\cos(\omega t)$ and the series on the right is the expansion for $\sin(\omega t)$:

$$\exp(\mathbf{A}t) = \mathbf{I}\cos(\omega t) + \mathbf{A}\frac{\sin(\omega t)}{\omega}$$

(iii)

The general solution to the initial value problem $\mathbf{x}' = \mathbf{A}\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$ is given by $\mathbf{x}(t) = \exp(\mathbf{A}t)\mathbf{x}_0$.

Using the expression derived in Part (ii):

$$\exp(\mathbf{A}t) = \begin{pmatrix} \cos(\omega t) & 0 \\ 0 & \cos(\omega t) \end{pmatrix} + \begin{pmatrix} 0 & \frac{\sin(\omega t)}{\omega} \\ -\omega \sin(\omega t) & 0 \end{pmatrix} = \begin{pmatrix} \cos(\omega t) & \frac{\sin(\omega t)}{\omega} \\ -\omega \sin(\omega t) & \cos(\omega t) \end{pmatrix}$$

To find the solution vector $\mathbf{x}(t)$, multiply this matrix by our initial condition vector $\mathbf{x}^0$:

$$\mathbf{x}(t) = \begin{pmatrix} \cos(\omega t) & \frac{\sin(\omega t)}{\omega} \\ -\omega \sin(\omega t) & \cos(\omega t) \end{pmatrix} \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}$$

Carrying out the matrix multiplication gives:

$$\mathbf{x}(t) = \begin{pmatrix} u_0 \cos(\omega t) + \frac{v_0}{\omega} \sin(\omega t) \\ -u_0 \omega \sin(\omega t) + v_0 \cos(\omega t) \end{pmatrix}$$

Since $x_1 = u$ and $x_2 = u'$, the solution to the original initial value problem is:

$$u(t) = u_0 \cos(\omega t) + \frac{v_0}{\omega} \sin(\omega t)$$

$$u'(t) = -u_0 \omega \sin(\omega t) + v_0 \cos(\omega t)$$

6

The eigenvalues are the roots of the characteristic polynomial, given by $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$:

$$\begin{vmatrix} 1-\lambda & 1 & 1 \\ 2 & 1-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} = -\lambda^3 + 3\lambda^2 - 4 = -(\lambda + 1)(\lambda - 2)^2 = 0$$

Thus, the eigenvalues are $\lambda_1 = -1$ (with algebraic multiplicity 1) and $\lambda_2 = 2$ (with algebraic multiplicity 2).

$$(\mathbf{A}+\mathbf{I})\boldsymbol{\xi}_1=\mathbf{0} \implies \begin{pmatrix}2 & 1 & 1\\ 2 & 2 & -1\\ 0 & -1 & 2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix} \implies \boldsymbol{\xi}_1=\begin{pmatrix}-3\\ 4\\ 2\end{pmatrix} \implies \mathbf{y}_1(t)=e^{-t}\begin{pmatrix}-3\\ 4\\ 2\end{pmatrix}$$

$$(\mathbf{A}-2\mathbf{I})\boldsymbol{\xi}_2=\mathbf{0} \implies \begin{pmatrix}-1 & 1 & 1\\ 2 & -1 & -1\\ 0 & -1 & -1\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix} \implies \boldsymbol{\xi}_2=\begin{pmatrix}0\\ -1\\ 1\end{pmatrix} \implies \mathbf{y}_2(t)=e^{2t}\begin{pmatrix}0\\ -1\\ 1\end{pmatrix}$$

Because the geometric multiplicity of $\lambda_2=2$ is 1 (which is strictly less than its algebraic multiplicity of 2), $\mathbf{A}$ is not diagonalizable. To find a generalized eigenvector $\boldsymbol{\eta}$ of rank 2 that satisfies $(\mathbf{A}-2\mathbf{I})\boldsymbol{\eta}=\boldsymbol{\xi}_2$:

$$\begin{pmatrix}-1 & 1 & 1\\ 2 & -1 & -1\\ 0 & -1 & -1\end{pmatrix}\begin{pmatrix}\eta_1\\ \eta_2\\ \eta_3\end{pmatrix}=\begin{pmatrix}0\\ -1\\ 1\end{pmatrix} \implies \boldsymbol{\eta}=\begin{pmatrix}-1\\ -1\\ 0\end{pmatrix}$$

Using the formula for repeated eigenvalues where algebraic multiplicity exceeds geometric multiplicity, the third linearly independent solution is:

$$\mathbf{y}_3(t)=(t\boldsymbol{\xi}_2+\boldsymbol{\eta})e^{2t}=\left[t\begin{pmatrix}0\\ -1\\ 1\end{pmatrix}+\begin{pmatrix}-1\\ -1\\ 0\end{pmatrix}\right]e^{2t}=\begin{pmatrix}-1\\ -t-1\\ t\end{pmatrix}e^{2t}$$

By the principle of superposition, the general solution is the linear combination of the fundamental set of solutions $(\mathbf{y}_1,\mathbf{y}_2,\mathbf{y}_3)$:

$$\mathbf{x}(t)=c_1\mathbf{y}_1(t)+c_2\mathbf{y}_2(t)+c_3\mathbf{y}_3(t)=c_1\begin{pmatrix}-3\\ 4\\ 2\end{pmatrix}e^{-t}+c_2\begin{pmatrix}0\\ -1\\ 1\end{pmatrix}e^{2t}+c_3\begin{pmatrix}-1\\ -t-1\\ t\end{pmatrix}e^{2t}$$

7

Since the matrix is lower triangular, the system is solved from top to bottom.

$$x_1'=x_1, \qquad x_1(0)=-1 \implies x_1(t)=-e^t.$$

$$x_2'=-4x_1+x_2=x_2+4e^t, \qquad x_2(0)=2 \implies x_2'-x_2=4e^t.$$

Using the integrating factor e^{-t},

$$(e^{-t}x_2)'=4,$$

hence

$$e^{-t}x_2=4t+C.$$

From $x_2(0)=2$: $C=2$,

$$x_2(t)=e^t(4t+2).$$

Finally,

$$x_3'=3x_1+6x_2+2x_3=e^t(24t+9)+2x_3$$

So

$$x_3'-2x_3=e^t(24t+9).$$

Let $x_3=e^ty$. Then

$$x_3'-2x_3=e^t(y'-y),$$

so

$$y'-y=24t+9.$$

A particular solution is $y_p=-24t-33$, and the homogeneous solution is $y_h=Ce^t$. Hence

$$y=Ce^t-24t-33,$$

therefore

$$x_3(t)=Ce^{2t}-(24t+33)e^t.$$

Using $x_3(0)=-30$,

$$C-33=-30 \quad \Rightarrow \quad C=3.$$

Thus

$$x_3(t)=3e^{2t}-(24t+33)e^t.$$

So the solution of the IVP is

$$\mathbf{x}(t)=e^t\begin{pmatrix}-1\\ 4t+2\\ 3e^t-24t-33\end{pmatrix}.$$

8

(i)

$$\mathbf{J}^2 = \mathbf{J}\mathbf{J} = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda^2 & 2\lambda & 1 \\ 0 & \lambda^2 & 2\lambda \\ 0 & 0 & \lambda^2 \end{pmatrix}$$

$$\mathbf{J}^3 = \mathbf{J}^2\mathbf{J} = \begin{pmatrix} \lambda^2 & 2\lambda & 1 \\ 0 & \lambda^2 & 2\lambda \\ 0 & 0 & \lambda^2 \end{pmatrix} \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda^3 & 3\lambda^2 & 3\lambda \\ 0 & \lambda^3 & 3\lambda^2 \\ 0 & 0 & \lambda^3 \end{pmatrix}$$

$$\mathbf{J}^4 = \mathbf{J}^3\mathbf{J} = \begin{pmatrix} \lambda^3 & 3\lambda^2 & 3\lambda \\ 0 & \lambda^3 & 3\lambda^2 \\ 0 & 0 & \lambda^3 \end{pmatrix} \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda^4 & 4\lambda^3 & 6\lambda^2 \\ 0 & \lambda^4 & 4\lambda^3 \\ 0 & 0 & \lambda^4 \end{pmatrix}$$

(ii)

Base Case ($n = 1$):

Substituting $n = 1$ into the formula yields:

$$\mathbf{J}^1 = \begin{pmatrix} \lambda^1 & 1 \cdot \lambda^0 & \frac{1(0)}{2}\lambda^{-1} \\ 0 & \lambda^1 & 1 \cdot \lambda^0 \\ 0 & 0 & \lambda^1 \end{pmatrix} = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

This matches $\mathbf{J}$, so the base case holds.

Inductive Step:

Assume that the formula holds for an arbitrary integer $k \geq 1$:

$$\mathbf{J}^k = \begin{pmatrix} \lambda^k & k\lambda^{k-1} & \frac{k(k-1)}{2}\lambda^{k-2} \\ 0 & \lambda^k & k\lambda^{k-1} \\ 0 & 0 & \lambda^k \end{pmatrix}$$

Multiply $\mathbf{J}^k$ by $\mathbf{J}$:

$$\mathbf{J}^{k+1} = \mathbf{J}^k\mathbf{J} = \begin{pmatrix} \lambda^k & k\lambda^{k-1} & \frac{k(k-1)}{2}\lambda^{k-2} \\ 0 & \lambda^k & k\lambda^{k-1} \\ 0 & 0 & \lambda^k \end{pmatrix} \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

Calculating the entries of the resulting matrix:

- **Row 1, Col 1:** $\lambda^k \cdot \lambda + k\lambda^{k-1} \cdot 0 + \cdots = \lambda^{k+1}$
- **Row 1, Col 2:** $\lambda^k \cdot 1 + k\lambda^{k-1} \cdot \lambda + 0 = (k+1)\lambda^k$
- **Row 1, Col 3:** $\lambda^k \cdot 0 + k\lambda^{k-1} \cdot 1 + \frac{k(k-1)}{2}\lambda^{k-2} \cdot \lambda = k\lambda^{k-1} + \frac{k^2-k}{2}\lambda^{k-1} = \frac{2k+k^2-k}{2}\lambda^{k-1} = \frac{(k+1)k}{2}\lambda^{k-1}$
- **Row 2, Col 2:** $0 \cdot 1 + \lambda^k \cdot \lambda + 0 = \lambda^{k+1}$
- **Row 2, Col 3:** $0 \cdot 0 + \lambda^k \cdot 1 + k\lambda^{k-1} \cdot \lambda = (k+1)\lambda^k$
- **Row 3, Col 3:** $0 + 0 + \lambda^k \cdot \lambda = \lambda^{k+1}$

Assembling the matrix yields:

$$\mathbf{J}^{k+1} = \begin{pmatrix} \lambda^{k+1} & (k+1)\lambda^k & \frac{(k+1)k}{2}\lambda^{k-1} \\ 0 & \lambda^{k+1} & (k+1)\lambda^k \\ 0 & 0 & \lambda^{k+1} \end{pmatrix}$$

This is exactly the assumed formula evaluated at $n = k + 1$. Therefore, by mathematical induction, the formula holds for all integers $n \geq 1$.

(iii)

The exponential of a 3×3 matrix with a single distinct eigenvalue λ can be efficiently evaluated using the Semisimple-Nilpotent (S-N) decomposition. Decompose the matrix as $\mathbf{J} = \mathbf{S} + \mathbf{N}$.

Define $\mathbf{S}$ as a multiple of the identity matrix and $\mathbf{N}$ as the remainder:

$$\mathbf{S} = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} = \lambda\mathbf{I}, \quad \mathbf{N} = \mathbf{J} - \mathbf{S} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Because $\mathbf{S}$ is a diagonal matrix, its exponential is straightforward: $e^{\mathbf{S}t} = e^{\lambda t}\mathbf{I}$.

For the nilpotent matrix $\mathbf{N}$:

$$\mathbf{N}^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{N}^3 = \mathbf{O}$$

Since $\mathbf{N}^3 = \mathbf{O}$, the Taylor series for $e^{\mathbf{N}t}$ truncates perfectly:

$$e^{\mathbf{N}t} = \mathbf{I} + \mathbf{N}t + \frac{1}{2}\mathbf{N}^2t^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & t & 0 \\ 0 & 0 & t \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & \frac{t^2}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

Because $\mathbf{S}$ is a scalar multiple of the identity matrix, it commutes with $\mathbf{N}$ (i.e., $\mathbf{SN} = \mathbf{NS}$). Therefore:

$$\exp(\mathbf{J}t) = e^{\mathbf{S}t}e^{\mathbf{N}t} = e^{\lambda t}\begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} & \frac{t^2}{2}e^{\lambda t} \\ 0 & e^{\lambda t} & te^{\lambda t} \\ 0 & 0 & e^{\lambda t} \end{pmatrix}$$

9

(i)

Let

$$A = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}.$$

Solve

$$\mathbf{x}' = A\mathbf{x}.$$

The characteristic polynomial is

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & -1 \\ 3 & -2-\lambda \end{vmatrix} = \lambda^2 - 1.$$

Hence the eigenvalues are

$$\lambda_1 = 1, \qquad \lambda_2 = -1.$$

For $\lambda_1 = 1$, an eigenvector is

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = -1$, an eigenvector is

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

So

$$\mathbf{x}_c(t) = c_1 e^{t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Split the forcing term:

$$\begin{pmatrix} e^t \\ t \end{pmatrix} = \begin{pmatrix} e^t \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ t \end{pmatrix}.$$

For $\begin{pmatrix} e^t \\ 0 \end{pmatrix}$:

Because e^t corresponds to the eigenvalue 1, the proposed form is

$$\mathbf{x}_{p1}(t) = e^t(\mathbf{a}t + \mathbf{b}).$$

Substituting gives

$$\mathbf{x}'_{p1} - A\mathbf{x}_{p1} = e^t\Big[(I - A)\mathbf{a}\,t + \mathbf{a} + (I - A)\mathbf{b}\Big].$$

To eliminate the t -term, it is required that

$$(I - A)\mathbf{a} = 0,$$

so $\mathbf{a}$ must be an eigenvector for $\lambda = 1$. Let

$$\mathbf{a} = \frac{3}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then solving

$$\mathbf{a} + (I - A)\mathbf{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

gives a convenient choice

$$\mathbf{b} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}.$$

Thus

$$\mathbf{x}_{p1}(t) = e^t\begin{pmatrix} \frac{3}{2}t \\ \frac{3}{2}t - \frac{1}{2} \end{pmatrix}.$$

For $\begin{pmatrix} 0 \\ t \end{pmatrix}$:

Try a polynomial vector

$$\mathbf{x}_{p2}(t) = \mathbf{c}t + \mathbf{d}.$$

Then

$$\mathbf{x}_{p2}' - A\mathbf{x}_{p2} = -A\mathbf{c}\,t + (\mathbf{c} - A\mathbf{d}).$$

Matching coefficients with $\begin{pmatrix}0\\ t\end{pmatrix}$, the system is

$$-A\mathbf{c} = \begin{pmatrix}0\\1\end{pmatrix}, \qquad \mathbf{c} - A\mathbf{d} = \begin{pmatrix}0\\0\end{pmatrix}.$$

Solving:

$$\mathbf{c} = \begin{pmatrix}1\\2\end{pmatrix}, \qquad \mathbf{d} = \begin{pmatrix}0\\-1\end{pmatrix}.$$

Hence

$$\mathbf{x}_{p2}(t) = \begin{pmatrix}t\\2t-1\end{pmatrix}.$$

So a particular solution is

$$\mathbf{x}_p(t) = e^t \begin{pmatrix}\frac{3}{2}t\\ \frac{3}{2}t - \frac{1}{2}\end{pmatrix} + \begin{pmatrix}t\\2t-1\end{pmatrix}.$$

Therefore,

$$\boxed{\mathbf{x}(t) = c_1 e^t \begin{pmatrix}1\\1\end{pmatrix} + c_2 e^{-t}\begin{pmatrix}1\\3\end{pmatrix} + e^t \begin{pmatrix}\frac{3}{2}t\\ \frac{3}{2}t - \frac{1}{2}\end{pmatrix} + \begin{pmatrix}t\\2t-1\end{pmatrix}}.$$

(ii)

Let

$$A = \begin{pmatrix}1 & \sqrt{3}\\ \sqrt{3} & -1\end{pmatrix}.$$

The characteristic polynomial is

$$\det(A - \lambda I) = \begin{vmatrix}1-\lambda & \sqrt{3}\\ \sqrt{3} & -1-\lambda\end{vmatrix} = \lambda^2 - 4.$$

Hence

$$\lambda_1 = 2, \qquad \lambda_2 = -2.$$

For $\lambda_1 = 2$, an eigenvector is

$$\mathbf{v}_1 = \begin{pmatrix}\sqrt{3}\\1\end{pmatrix}.$$

For $\lambda_2 = -2$, an eigenvector is

$$\mathbf{v}_2 = \begin{pmatrix}1\\-\sqrt{3}\end{pmatrix}.$$

Thus

$$\mathbf{x}_c(t) = c_1 e^{2t}\begin{pmatrix}\sqrt{3}\\1\end{pmatrix} + c_2 e^{-2t}\begin{pmatrix}1\\-\sqrt{3}\end{pmatrix}.$$

Split the forcing term:

$$\begin{pmatrix}e^t\\ \sqrt{3}e^{-t}\end{pmatrix} = \begin{pmatrix}e^t\\0\end{pmatrix} + \begin{pmatrix}0\\ \sqrt{3}e^{-t}\end{pmatrix}.$$

For $\begin{pmatrix}e^t\\0\end{pmatrix}$:

Since 1 is not an eigenvalue, the proposed form is

$$\mathbf{x}_{p1}(t) = \mathbf{a}e^t.$$

Then

$$\mathbf{x}_{p1}' - A\mathbf{x}_{p1} = (I - A)\mathbf{a}\,e^t.$$

Solving for $\mathbf{a}$:

$$(I - A)\mathbf{a} = \begin{pmatrix}1\\0\end{pmatrix}.$$

Now

$$I - A = \begin{pmatrix}0 & -\sqrt{3}\\ -\sqrt{3} & 2\end{pmatrix},$$

and solving gives

$$\mathbf{a} = \begin{pmatrix} -\frac{2}{3} \\ -\frac{\sqrt{3}}{3} \end{pmatrix}.$$

Hence

$$\mathbf{x}_{p1}(t) = e^{t} \begin{pmatrix} -\frac{2}{3} \\ -\frac{\sqrt{3}}{3} \end{pmatrix}.$$

For $\begin{pmatrix} 0 \\ \sqrt{3}e^{-t} \end{pmatrix}$:

Since -1 is not an eigenvalue, the proposed form is

$$\mathbf{x}_{p2}(t) = \mathbf{b}e^{-t}.$$

Then

$$\mathbf{x}_{p2}' - A\mathbf{x}_{p2} = (-I - A)\mathbf{b}\, e^{-t}.$$

Solving for $\mathbf{b}$:

$$(-I - A)\mathbf{b} = \begin{pmatrix} 0 \\ \sqrt{3} \end{pmatrix},$$

which is equivalent to

$$(I + A)\mathbf{b} = \begin{pmatrix} 0 \\ -\sqrt{3} \end{pmatrix}.$$

Since

$$I + A = \begin{pmatrix} 2 & \sqrt{3} \\ \sqrt{3} & 0 \end{pmatrix},$$

it is found that

$$\mathbf{b} = \begin{pmatrix} -1 \\ \frac{2\sqrt{3}}{3} \end{pmatrix}.$$

Therefore

$$\mathbf{x}_{p2}(t) = e^{-t} \begin{pmatrix} -1 \\ \frac{2\sqrt{3}}{3} \end{pmatrix}.$$

So

$$\mathbf{x}_{p}(t) = e^{t} \begin{pmatrix} -\frac{2}{3} \\ -\frac{\sqrt{3}}{3} \end{pmatrix} + e^{-t} \begin{pmatrix} -1 \\ \frac{2\sqrt{3}}{3} \end{pmatrix}.$$

Hence,

$$\boxed{\mathbf{x}(t) = c_1 e^{2t} \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} + e^{t} \begin{pmatrix} -\frac{2}{3} \\ -\frac{\sqrt{3}}{3} \end{pmatrix} + e^{-t} \begin{pmatrix} -1 \\ \frac{2\sqrt{3}}{3} \end{pmatrix}}.$$

(iii)

Let

$$A = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix}.$$

Because the forcing term involves $\csc t, \sec t$, variation of parameters is the natural method here.

The characteristic polynomial is

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & -5 \\ 1 & -2 - \lambda \end{vmatrix} = \lambda^2 + 1.$$

So the eigenvalues are

$$\lambda = \pm i.$$

A very convenient observation is

$$A^2 = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I.$$

Hence, using the matrix-exponential formulation,

$$e^{At} = I\cos t + A\sin t.$$

Therefore,

$$e^{At} = \begin{pmatrix} \cos t + 2\sin t & -5\sin t \\ \sin t & \cos t - 2\sin t \end{pmatrix}.$$

So the homogeneous solution can be written as

$$\mathbf{x}_c(t) = c_1 \begin{pmatrix} \cos t + 2\sin t \\ \sin t \end{pmatrix} + c_2 \begin{pmatrix} -5\sin t \\ \cos t - 2\sin t \end{pmatrix}.$$

Take

$$\Phi(t) = e^{At} = I\cos t + A\sin t.$$

Since $A^2 = -I$,

$$\Phi^{-1}(t) = e^{-At} = I\cos t - A\sin t.$$

A particular solution is

$$\mathbf{x}_p(t) = \Phi(t)\int \Phi^{-1}(t)\mathbf{g}(t)\,dt, \qquad \mathbf{g}(t) = \begin{pmatrix} \csc t \\ \sec t \end{pmatrix}.$$

Now

$$A\mathbf{g}(t) = \begin{pmatrix} 2\csc t - 5\sec t \\ \csc t - 2\sec t \end{pmatrix},$$

so

$$\Phi^{-1}(t)\mathbf{g}(t) = \cos t \begin{pmatrix} \csc t \\ \sec t \end{pmatrix} - \sin t \begin{pmatrix} 2\csc t - 5\sec t \\ \csc t - 2\sec t \end{pmatrix}.$$

Thus

$$\Phi^{-1}(t)\mathbf{g}(t) = \begin{pmatrix} \cot t - 2 + 5\tan t \\ 2\tan t \end{pmatrix}.$$

Integrating componentwise:

$$\int \begin{pmatrix} \cot t - 2 + 5\tan t \\ 2\tan t \end{pmatrix} dt = \begin{pmatrix} \ln(\sin t) - 2t - 5\ln|\cos t| \\ -2\ln|\cos t| \end{pmatrix}.$$

Here $\frac{\pi}{2} < t < \pi$, so $\sin t > 0$ and $\cos t < 0$, hence $|\cos t|$ is appropriate.

Therefore,

$$\mathbf{x}_p(t) = \begin{pmatrix} \cos t + 2\sin t & -5\sin t \\ \sin t & \cos t - 2\sin t \end{pmatrix} \begin{pmatrix} \ln(\sin t) - 2t - 5\ln|\cos t| \\ -2\ln|\cos t| \end{pmatrix}.$$

The particular solution can be left in this form, or expanded:

$$\mathbf{x}_p(t) = \begin{pmatrix} (\cos t + 2\sin t)(\ln(\sin t) - 2t) - 5\cos t\,\ln|\cos t| \\ \sin t\big(\ln(\sin t) - 2t - 3\ln|\cos t|\big) - 2\cos t\,\ln|\cos t| \end{pmatrix}.$$

Hence,

$$\boxed{\mathbf{x}(t) = c_1 \begin{pmatrix} \cos t + 2\sin t \\ \sin t \end{pmatrix} + c_2 \begin{pmatrix} -5\sin t \\ \cos t - 2\sin t \end{pmatrix} + \begin{pmatrix} (\cos t + 2\sin t)(\ln(\sin t) - 2t) - 5\cos t\,\ln|\cos t| \\ \sin t\big(\ln(\sin t) - 2t - 3\ln|\cos t|\big) - 2\cos t\,\ln|\cos t| \end{pmatrix}, \quad \frac{\pi}{2} < t < \pi}.$$